

PROGRAMMING ASSIGNMENT 3

The Aetheria Project: Architecting the City of Tomorrow

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Programming Language: Java (OpenJDK 11)

Due Date: **Saturday, 02.05.2026 (23:59:59)**

1 Background Story

You have been appointed as the Chief Systems Architect for Aetheria, the world's first fully automated smart city. Aetheria is not just a collection of buildings; it is a living organism of power grids, data fiber, and interdependent control systems.

Your predecessor left behind a chaotic blueprint. Your job is to use the provided Graph and Digraph APIs to organize the city's infrastructure. The city council demands efficiency, safety, and a perfect boot-up sequence.

2 The Blueprints

2.1 The Power Grid Dilemma

Aetheria consists of 13 districts (indexed 0-12). To save costs, the council wants to lay the minimum amount of high-voltage cable possible while ensuring that every district is reachable from every other district.

- **The Mission:** First, identify if the current city districts are already physically connected or if they are split into isolated "islands".
- **Optimization:** Design a module that finds the "most efficient" way to link all districts so that the total cabling cost is minimized.
- **API Note:** Your solution must use the Graph class and iterate via the `adj(v)` method. The city map with all edge weights and the highlighted MST is shown in Figure 1.

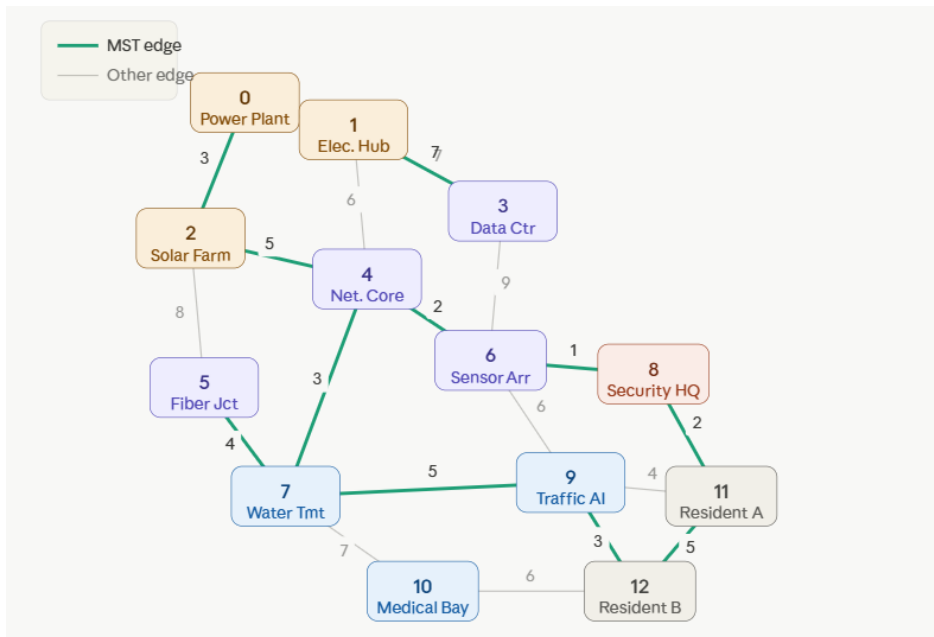


Figure 1. The city map showing all 13 districts. Teal edges are the MST (minimum cable); gray edges are the remaining connections. Edge weights (cable costs) are labeled inline.

2.2 The Master Boot Sequence

Aetheria's core systems—Water, Electricity, Security, and the Data Center—cannot all start at once. For example, the Security system requires the Data Center to be online first, and the Data Center requires Electricity.

- The Mission: You must generate a valid Opening Protocol (a linear order) so that every system is activated only after its requirements are met.
- The Critical Failure: If there is a "Circular Reference" in the dependencies (e.g., A needs B, B needs C, and C needs A), the city will enter a Deadlock. Your code must detect these cycles and trigger a "System Infeasible" alert. Figure 2 illustrates both the valid boot ordering and the deadlock scenario.

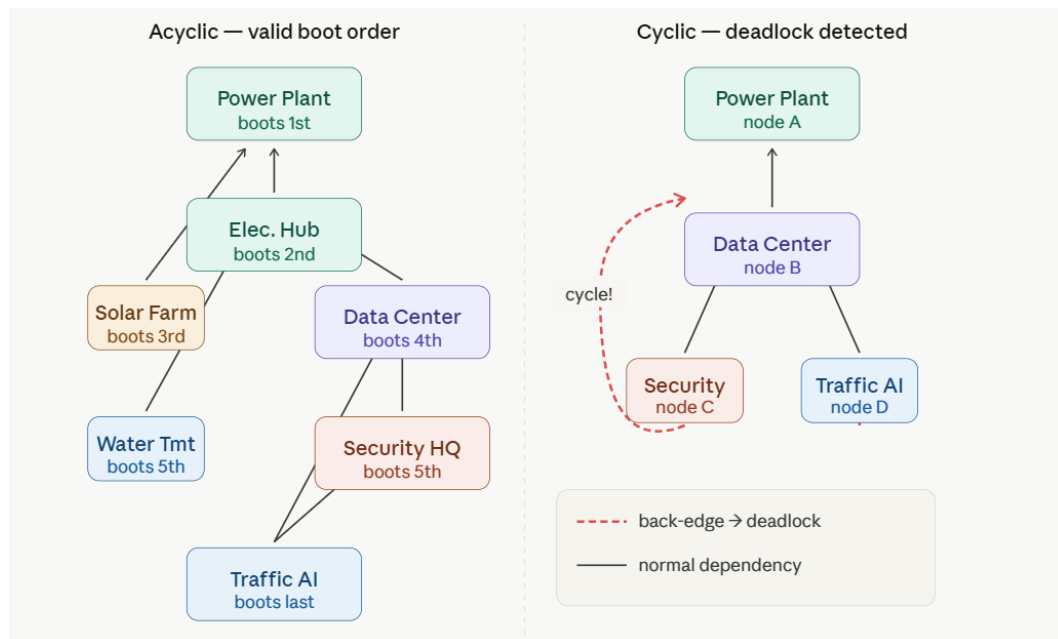


Figure 2. Left: the valid dependency DAG with each node labeled by its boot position. Right: the same graph with a back-edge (dashed red) that creates a cycle and triggers the deadlock alert.

2.3 Secure Communication Hubs

The city's internal data network uses one-way fiber optic lines for security.

- The Mission: The council needs to identify "High-Interaction Zones." These are groups of stations where any station in the group can send a message to any other station in the same group and receive a reply back. The directed network with both High-Interaction Zones highlighted is shown in Figure 3

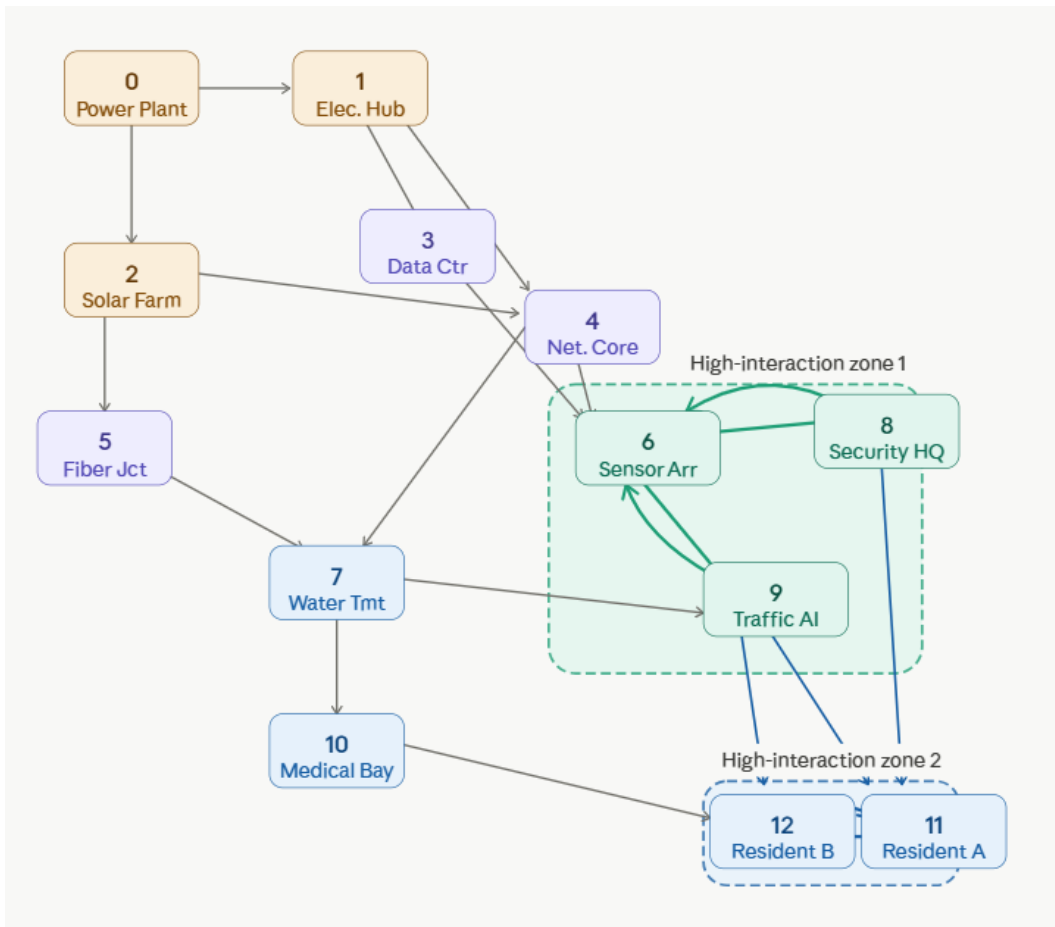


Figure 3. The directed network with the two High-Interaction Zones highlighted (dashed outlines). Teal bidirectional arrows inside the zones show that every station can reach every other station and receive a reply.

2.4 The Bio-Leak Containment Protocol

In the event of a simulated emergency (like a digital virus or a physical leak) at a specific station 's', we must know the scope of the danger.

- Task 1: List every station that the leak could eventually reach from the source station(s).
- Task 2: For each reachable station, find the minimum number of connections the leak must travel through to reach it. Group and print stations by this distance. Figure 4 illustrates the expected output structure for both tasks. The underlying Bag-based adjacency list representation used by the Graph API is shown in Figure 5.

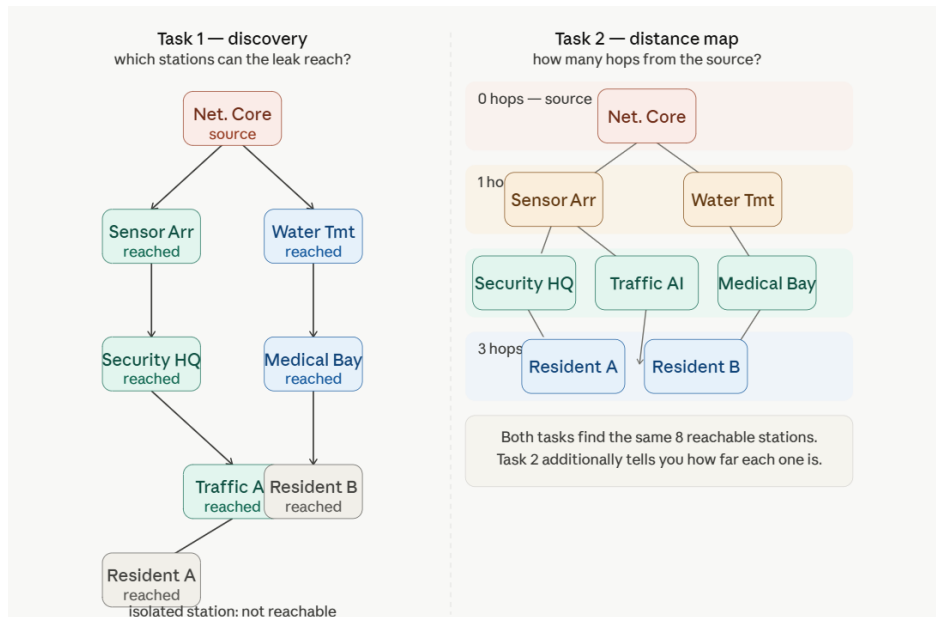


Figure 4. Task 1: Discovery and Task 2: Distance Map

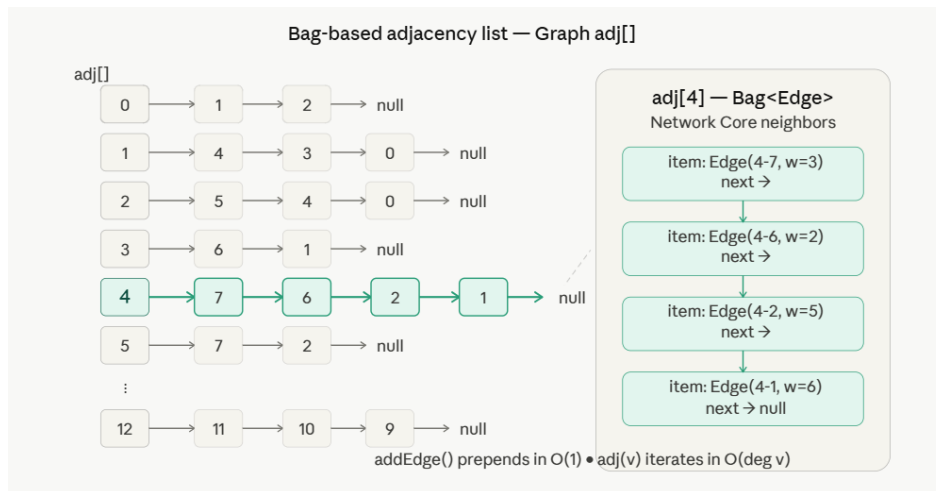


Figure 5. The full array with linked-list Bags, with vertex 4 (Network Core), exploded to show each Edge node with its next pointer.

3 Complete Example

3.1 Input

A part of the XML fragment is provided below for clarity.

```
<data>
  <districts>
    <district>
      <id>D0</id>
      <name>Power Plant</name>
      <links>
        <link>
          <target>D1</target>
          <cost>4</cost>
        </link>
        <link>
          <target>D2</target>
          <cost>3</cost>
        </link>
      </links>
    </district>
    <district>
      <id>D1</id>
      <name>Electricity Hub</name>
      <links>
        <link>
          <target>D3</target>
          <cost>7</cost>
        </link>
        <link>
          <target>D4</target>
          <cost>6</cost>
        </link>
      </links>
    </district>
    <district>
      <id>D2</id>
      <name>Solar Farm</name>
      <links>
        <link>
          <target>D4</target>
          <cost>5</cost>
        </link>
        <link>
          <target>D5</target>
          <cost>8</cost>
        </link>
      </links>
    </district>
    <district>
      <id>D3</id>
      <name>Data Center</name>
      <links>
        <link>
          <target>D6</target>
          <cost>9</cost>
        </link>
      </links>
    </district>
    <district>
      <id>D4</id>
      <name>Network Core</name>
      <links>
        <link>
          <target>D6</target>
          <cost>2</cost>
        </link>
      </links>
    </district>
  </districts>
</data>
```

3.2 Output

The expected output is provided below for clarity.

```
AETHERIA CITY - Full System Boot Report

PART A | Power Grid

City fully connected? YES ✓

Minimum Spanning Tree (12 cables):
  Sensor Array      ↗ Security HQ      cost: 1
  Network Core      ↗ Sensor Array     cost: 2
  Security HQ       ↗ Residential A    cost: 2
  Power Plant       ↗ Solar Farm       cost: 3
  Network Core      ↗ Water Treatment  cost: 3
  Traffic AI Centre ↗ Residential B    cost: 3
  Power Plant       ↗ Electricity Hub  cost: 4
  Fiber Junction    ↗ Water Treatment  cost: 4
  Traffic AI Centre ↗ Residential A    cost: 4
  Solar Farm        ↗ Network Core     cost: 5
  Medical Bay       ↗ Residential B    cost: 6
  Electricity Hub   ↗ Data Center       cost: 7
Total minimum cable cost: 44 units

PART B | Master Boot Sequence (valid order)

=== Part B: Master Boot Sequence ===
Valid boot sequence found:
  Step 1: Traffic AI (id=6)
  Step 2: Water Treatment (id=5)
  Step 3: Security HQ (id=4)
  Step 4: Data Center (id=3)
  Step 5: Solar Farm (id=2)
  Step 6: Electricity Hub (id=1)
  Step 7: Power Plant (id=0)

--- Introducing circular dependency (deadlock demo) ---
=== Part B: Master Boot Sequence ===
[Cycle detected] Edge 1 → 0 creates a circular dependency.
SYSTEM INFEASIBLE: Circular reference (deadlock) detected!

PART C | Secure Communication Hubs (SCCs)

=== Part C: Secure Communication Hubs ===
Total SCCs found: 10
SCC-0 [HIGH-INTERACTION ZONE]: {Residential A, Residential B}
SCC-1: {Medical Bay}
SCC-2 [HIGH-INTERACTION ZONE]: {Sensor Array, Security HQ, Traffic AI Centre}
SCC-3: {Water Treatment}
SCC-4: {Fiber Junction}
SCC-5: {Network Core}
SCC-6: {Data Center}
SCC-7: {Solar Farm}
SCC-8: {Electricity Hub}
SCC-9: {Power Plant}
High-Interaction Zones: 2

PART D | Bio-Leak Containment Protocol

Leak origin: Network Core (id=4)

=== Part D, Task 1 Network Core(4) ===
Stations at risk (8):
  Network Core(4)
  Water Treatment(7)
  Medical Bay(10)
  Residential B(12)
  Residential A(11)
  Traffic AI Centre(9)
  Sensor Array(6)
  Security HQ(8)

=== Part D, Task 2 Network Core(4) ===
Evacuation priority layers:
  Layer 0 (source) : Network Core(4)
  Layer 1 (priority 1): Sensor Array(6), Water Treatment(7)
  Layer 2 (priority 2): Security HQ(8), Traffic AI Centre(9), Medical Bay(10)
  Layer 3 (priority 3): Residential A(11), Residential B(12)
```

4 Grading Policy

- The Power Grid Tests → 20%
- The master boot sequence tests → 20%
- The secure communication Hubs tests → 20%
- The bio-leak protocol tests 20%
- Output tests → 20%

5 Important Notes

- Do not miss the deadline: **Saturday, 02.05.2026 (23:59:59)**.
- Save all your work until the assignment is graded.
- Usage of any external libraries is forbidden (The use of javax.xml is considered part of the standard Java library and does not violate the 'no external libraries' constraint specified in the assignment).
- Use the provided [starter code](#) here.
- The assignment solution you submit must be your original, individual work. Duplicate or similar assignments are both going to be considered as cheating.
- You can ask your questions via Piazza (<https://piazza.com/hacettepe.edu.tr/spring2026/bbm204>), and you are supposed to be aware of everything discussed on Piazza.
- You will submit your work via <https://submit.cs.hacettepe.edu.tr/> with the file hierarchy given below:

b<studentID>.zip

```
|
├─ AetheriaCity.java <FILE>
|
├─ PowerGrid.java <FILE>
|
├─ BootSequence.java <FILE>
|
├─ CommHubs.java <FILE>
|
└─ BioLeakContainment.java <FILE>
```

- The AetheriaCity is the main class that contains the main method. The main class and all other classes should be placed directly (no subfolders) into a zip file named **b<studentID>.zip**.
- The other needed files to compile: Bag.java, Edge.java, Graph.java, Digraph.java will be given in the starter code file.

- Feel free to create subdirectories if necessary, but they should also be in **b<studentID>.zip** directory.
- This file hierarchy must be zipped before submission (not .rar, only .zip files are supported).

Academic Integrity Policy

All work on assignments **must be done individually**. You are encouraged to discuss the given assignments with your classmates, but these discussions should be carried out in an abstract way. That is, discussions related to a particular solution to a specific problem (either in actual code or in pseudocode) **will not be tolerated**. In short, turning in someone else's work (including work available on the internet), in whole or in part, as your own will be considered as a **violation of academic integrity**. Please note that the former condition also holds for online/AI sources. Make use of them responsibly, refrain from generating the code you are asked to implement. Remember that we also have access to such tools, making it easier to detect such cases.



The submissions will be subjected to a similarity check. Any submissions that fail the similarity check will not be graded and will be reported to the ethics committee as a case of academic integrity violation, which may result in suspension of the involved students.